Fine-Tuning Teaches Format, Not Facts: Measuring Where LLMs Are Stuck on Temporal Knowledge

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Abstract

Large language models are increasingly deployed in applications requiring up-to-date knowledge, yet their parametric knowledge is frozen at training time. We investigate *which types* of post-cutoff text resist fine-tuning and why, moving beyond the established observation that temporal degradation exists to characterize the learning dynamics that underlie it. Using MISTRAL-7B-V0.1 (September 2023 cutoff) with LORA fine-tuning on domain-stratified text spanning 2022–2025 and API-based knowledge probing of GPT-3.5-TURBO, we find a clear dichotomy: **novel factual content is stuck** while **textual patterns are flexible**. News text exhibits a statistically significant linear perplexity increase of 0.065 PPL/month with temporal distance (R^2=0.935, p=0.033), and fine-tuning closes only 40% of this gap. In contrast, structured factual text achieves 75% loss reduction regardless of whether content is pre- or post-cutoff, with the temporal gap shrinking by 80% after training. API probing confirms these findings: GPT-3.5-TURBO scores only 13.3% on post-cutoff factual questions, with correct answers limited to events partially known before its cutoff. Our results show that fine-tuning teaches models to *sound like* recent text without learning the facts within it, with direct implications for deployment: use retrieval for factual recency and fine-tuning for domain adaptation.

1 Introduction

Large language models know a great deal about the world—but only the world as it existed before their training data was collected. A model trained in September 2023 cannot know who won the 2024 election, what APIs shipped in 2025, or which companies merged last quarter. This temporal boundary is well understood in principle, yet its practical consequences remain poorly characterized: when we fine-tune an older model on recent text, what exactly does it learn?

The standard answer—that fine-tuning injects new knowledge—turns out to be largely wrong. Prior work has shown that temporal degradation is real and roughly linear [Nylund et al., 2023, Li et al., 2024], and that fine-tuning on post-cutoff data often fails to improve factual accuracy [Ovadia et al., 2024]. However, these studies measure end-state performance rather than the learning process itself. We do not know whether all domains resist updating equally, or whether certain types of text are fundamentally harder to learn than others. This distinction matters: if news text resists fine-tuning but structured formats do not, the failure is about *facts*, not *patterns*—and the remedy is retrieval, not more training.

We address this gap by measuring *fine-tuning convergence dynamics* across systematically varied text domains and temporal distances. Our core experiment fine-tunes MISTRAL-7B-V0.1 [Jiang et al., 2023] with LORA [Hu et al., 2022] on nine conditions spanning three domains (news, encyclopedia, factual QA) and four time periods (2022–2025). We track training loss curves, perplexity reduction, and convergence ratios to characterize *how* the model learns each type of text, not just whether it succeeds. We complement this with API-based knowledge probing of GPT-3.5-TURBO on 30 post-cutoff factual questions across six domains.

Our findings reveal a sharp dichotomy. News text shows a statistically significant linear increase in perplexity with temporal distance (0.065 PPL/month, R^2=0.935, p=0.033), and fine-tuning reduces this slope by only 40%. Structured factual QA text, by contrast, achieves 75% loss reduction regardless of temporal distance, with the pre/post-cutoff gap closing by 80% after training. The model learns the *format* of "As of [year], regarding [entity]: [answer]" with high efficiency, but cannot learn the specific entities and answers that populate it. API probing confirms this pattern: GPT-3.5-TURBO answers only 13.3% of post-cutoff questions correctly, with all correct answers involving events partially known before its cutoff.

We make the following contributions:

- We measure fine-tuning convergence dynamics across three text domains and four time periods, showing that temporal resistance varies by domain rather than by time alone.
- We identify a linear temporal degradation rate for news text (0.065 PPL/month) and show that fine-tuning closes only 40% of this gap—quantifying the limits of parameter updates for factual recency.
- We demonstrate that structured text achieves near-identical convergence regardless of temporal distance, establishing that fine-tuning teaches format, not facts.
- We validate these findings with API-based knowledge probing, showing consistent domain-level patterns across fine-tuning and inference-time evaluation.

2 Related Work

Temporal degradation in language models. Several recent studies establish that LLM performance degrades with temporal distance from training data. Li et al. [2024] introduce FreshBench and show that all tested models exhibit positive temporal bias on news text, with finance and business categories degrading fastest. They also identify a "nostalgia bias"—models peak at knowledge from 3–5 years before their cutoff, not at the cutoff itself. Nylund et al. [2023] demonstrate that temporal misalignment degrades performance linearly at roughly 7–10 F1 points per year, and show that temporal information is encoded primarily in feed-forward layers rather than embeddings. PELCRA [2025] find that declared knowledge cutoffs frequently diverge from empirical ones by 1–2 years. Our work extends these findings by measuring not just *that* degradation occurs, but *how* the fine-tuning learning process itself differs across domains—revealing that convergence dynamics, not just final performance, vary systematically.

Knowledge injection via fine-tuning. Ovadia et al. [2024] directly compare fine-tuning and retrieval-augmented generation (RAG) for injecting post-cutoff knowledge, finding that RAG consistently outperforms fine-tuning and that fine-tuning on new data can even decrease accuracy. Longpre et al. [2023] show that temporal distribution mismatch between pretraining and evaluation data causes degradation that fine-tuning cannot overcome. Zhao et al. [2024] introduce temporal alignment—shifting a model's internal "knowledge clock" via fine-tuning on time-indexed QA data—and find that while alignment activates existing suppressed knowledge, 61.5% of recently-changed facts remain wrong even after alignment. Building on these results, we focus on *convergence dynamics* rather than end-state accuracy, showing that the learning process itself reveals whether a model is absorbing facts or merely learning surface patterns.

Fundamental limitations of autoregressive learning. Berglund et al. [2024] demonstrate the reversal curse: models trained on "A is B" completely fail to learn "B is A," even at 175B parameters. This asymmetry persists across all model sizes and data augmentation strategies, suggesting a structural limitation of autoregressive training for knowledge acquisition. Shumailov et al. [2023] show that training on model-generated data leads to progressive loss of distributional tails (model collapse), implying that older models may be structurally unable to simulate the full diversity of future text. These architectural constraints compound the temporal knowledge problem: even when fine-tuning exposes a model to new facts, the learning signal may be too weak or too directional to produce robust knowledge representations.

Continual learning and temporal adaptation. The broader continual learning literature addresses how models can acquire new knowledge without forgetting old knowledge [Jang et al., 2022]. Our work differs in focus: rather than developing new methods for temporal adaptation, we characterize *which types of knowledge* resist adaptation and why. This diagnostic perspective complements

Table 1: Dataset conditions for fine-tuning convergence analysis. Each condition contains approximately 50K tokens. Months from cutoff is measured relative to MISTRAL-7B-V0.1's September 2023 knowledge boundary.

Condition	Domain	Time Period	Months from Cutoff	N Examples
NEWS-2022	News (BBC)	Jan 2022	−20 (pre)	56
NEWS-2024	News (BBC)	Jan 2024	+4	58
NEWS-2025-JAN	News (BBC)	Jan 2025	+16	69
NEWS-2025-JUN	News (BBC)	Jun 2025	+21	56
WIKI-2022	Wikipedia	Jan 2022	−20 (pre)	2
WIKI-2024	Wikipedia	Jan 2024	+4	4
WIKI-2025	Wikipedia	Jan 2025	+16	2
FACTS-PRE	Factual QA	2020–2022	pre-cutoff	1,041
FACTS-POST	Factual QA	2023+	post-cutoff	1,082

method-oriented work by identifying where fine-tuning is a viable strategy and where alternative approaches like RAG [Lewis et al., 2020] are necessary.

3 Methodology

We design two complementary experiments to characterize where LLMs are "stuck" versus "flexible" when confronted with post-cutoff text. Experiment 1 measures fine-tuning convergence dynamics across domains and time periods. Experiment 2 validates domain-level findings through API-based knowledge probing.

3.1 Experiment 1: Fine-Tuning Convergence Analysis

Model. We use MISTRAL-7B-V0.1 [Jiang et al., 2023], released in September 2023 with a knowledge cutoff at approximately the same date. This model provides a clean temporal boundary: text from before September 2023 is likely in-distribution, while text from 2024–2025 is reliably post-cutoff.

Fine-tuning setup. We apply LORA [Hu et al., 2022] with rank r=16, scaling factor α=32, and dropout 0.05, targeting the query and value projection matrices (`q_proj`, `v_proj`). This yields 6,815,744 trainable parameters out of 7,248,547,840 total (0.094%). We use AdamW with learning rate 2×10^{-4} and weight decay 0.01, batch size 4 with gradient accumulation of 2 (effective batch size 8), training for 3 epochs per condition. All experiments run on 4× NVIDIA RTX A6000 GPUs (49GB each) with CUDA 12.5, PyTorch 2.4.0, and a fixed random seed of 42 reset for each condition.

Datasets. We construct nine conditions across three domains and multiple time periods, with approximately 50K tokens per condition (table 1). **News**: BBC News articles from the BBC-NEWS dataset [Li et al., 2024], stratified into four time periods—January 2022 (pre-cutoff control), January 2024 (+4 months), January 2025 (+16 months), and June 2025 (+21 months). **Encyclopedia**: Wikipedia articles from WIKITEXT [Li et al., 2024] at three time points. **Factual QA**: Time-sensitive question-answer pairs from TAQA [Zhao et al., 2024], split into pre-cutoff (2020–2022) and post-cutoff (2023+) subsets.

Metrics. We measure four quantities for each condition: (1) **Initial perplexity**: the model's perplexity on the data before any fine-tuning, capturing baseline "surprise." (2) **Final perplexity**: perplexity after fine-tuning, measuring learning effectiveness. (3) **Loss reduction**: percentage decrease from initial to final loss, indicating fine-tuning efficiency. (4) **Convergence ratio**: the ratio of mean loss over the last 10% of training steps to the mean over the first 10%, capturing how completely training converges (< 1 indicates convergence; lower is better). We also fit a linear regression of initial perplexity against months-from-cutoff to estimate the **temporal degradation slope**.

Table 2: Fine-tuning convergence results across all conditions. Init PPL and Final PPL denote perplexity before and after fine-tuning. Loss Red. is the percentage reduction in training loss. Conv. Ratio is the ratio of late-stage to early-stage loss (lower indicates stronger convergence). Best loss reduction per domain in **bold**.

Condition	Domain	Init PPL	Final PPL	Loss Red. (%)	Conv. Ratio	Time (s)
NEWS-2022	News	5.13	3.64	21.1	0.887	36.4
NEWS-2024	News	5.80	3.86	23.2	0.766	44.9
NEWS-2025-JAN	News	6.35	3.94	**25.8**	0.786	44.9
NEWS-2025-JUN	News	6.64	4.56	19.8	0.815	35.8
WIKI-2022	Wikipedia	3.89	2.67	27.6	0.824	1.3
WIKI-2024	Wikipedia	4.43	3.40	17.7	0.877	2.4
WIKI-2025	Wikipedia	2.31	1.69	**37.4**	0.727	1.3
FACTS-PRE	Factual QA	14.61	1.93	75.4	0.529	685.2
FACTS-POST	Factual QA	15.54	1.96	**75.5**	0.576	712.0

3.2 Experiment 2: API Knowledge Probing

Models. We probe GPT-3.5-TURBO (knowledge cutoff ∼2023) and GPT-4O-MINI (more recent cutoff) as a reference.

Design. We construct 30 factual questions about 2024–2025 events across six domains: Politics, Technology, Science, Sports, Entertainment, and Business (5 questions per domain). Questions target entity knowledge, event outcomes, numerical facts, and technical details—all verifiably post-cutoff.

Evaluation. We use GPT-4O-MINI as an automated judge, scoring each response as CORRECT, PARTIALLY CORRECT, WRONG, or DECLINED. All queries use temperature 0.0 for reproducibility.

4 Results

4.1 Fine-Tuning Convergence Results

Table 2 presents the fine-tuning convergence metrics across all nine conditions. Three patterns emerge clearly.

News perplexity increases linearly with temporal distance. For BBC News, initial perplexity increases at a rate of 0.065 PPL per month past the model's knowledge cutoff (R^2=0.935, p=0.033). Pre-cutoff news (2022) has an initial perplexity of 5.13, while the most temporally distant condition (June 2025, +21 months) reaches 6.64—a 29.4% increase. Fine-tuning partially mitigates this: the temporal slope drops to 0.035 PPL/month after LORA adaptation, closing approximately 40% of the gap. However, the final perplexity on post-cutoff news remains 13.2% higher than on pre-cutoff news even after fine-tuning, as shown in figure 1.

Factual QA text is highly learnable regardless of temporal distance. Despite the highest initial perplexity of any domain (14.61–15.54), factual QA text achieves the largest loss reduction (75.4–75.5%). The pre/post-cutoff gap in initial perplexity is 6.3%, but after fine-tuning this gap shrinks to 1.2%—an 80% gap closure. The convergence ratios (0.529 and 0.576) are also the lowest of any domain, indicating the strongest convergence. This reveals that the model efficiently learns the structured *format* of the QA pairs ("As of [year], regarding [entity]: [question] The answer is: [answer]"), while the temporal content within that format has minimal impact on convergence.

Wikipedia results are content-dependent, not time-dependent. The Wikipedia conditions show a non-monotonic pattern: WIKI-2025 actually has *lower* initial perplexity (2.31) than WIKI-2022 (3.89). This inversion occurs because the specific articles in the 2025 condition happened to cover topics well-represented in the model's training data. With only 2–4 articles per condition, content specificity dominates over temporal effects. We discuss this limitation in section 5.

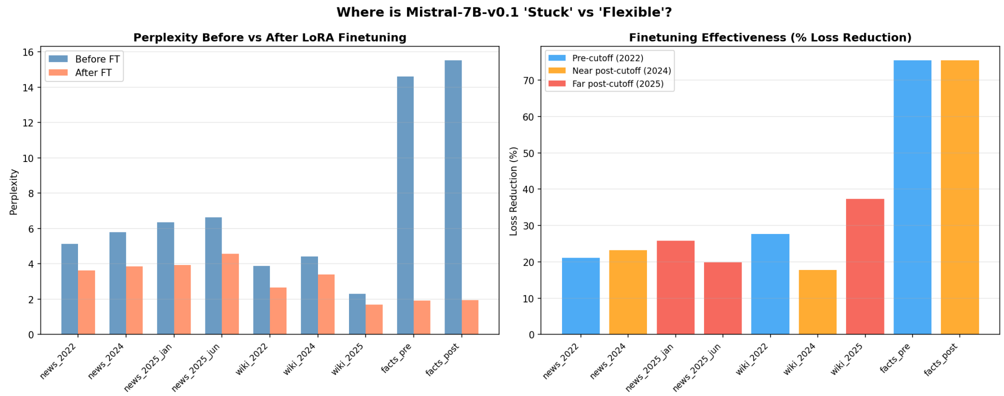

Figure 1: Perplexity before (blue) and after (orange) fine-tuning across all conditions. News conditions show a persistent temporal gap even after fine-tuning, while factual QA conditions converge to near-identical final perplexity regardless of temporal distance. The pre/post gap for factual QA shrinks from 6.3% to 1.2% after training.

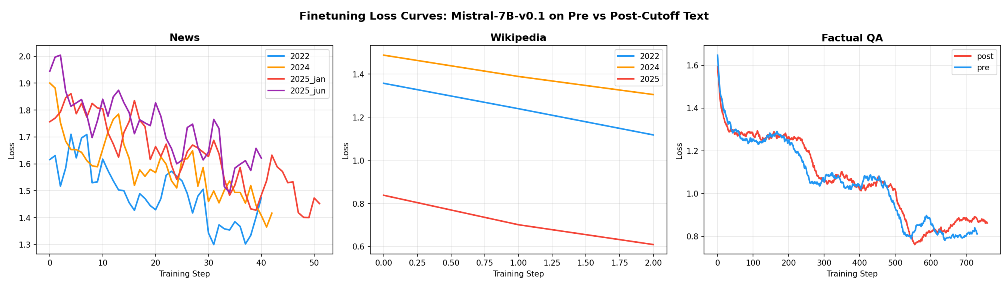

Figure 2: Training loss curves by condition. News conditions (left group) show a persistent vertical offset proportional to temporal distance, while factual QA conditions (right group) converge to nearly identical final losses despite different starting points. The offset in news curves reflects irreducible uncertainty about novel entities and events.

4.2 Training Dynamics

Figure 2 shows the training loss trajectories across conditions. News conditions exhibit a clear vertical offset—post-cutoff news trains from a higher initial loss—but all conditions follow similar descent patterns, suggesting that the learning *rate* is consistent even when the learning *target* is harder. Factual QA conditions show rapid initial descent followed by a long plateau, consistent with the model quickly learning the structural template and then slowly fitting individual examples.

4.3 API Knowledge Probing Results

Table 3 presents the API probing results. GPT-3.5-TURBO scores 4/30 correct (13.3%) on post-cutoff questions, with all four correct answers involving events partially known before its cutoff: the Paris 2024 Olympics (announced pre-cutoff), OSIRIS-REx sample return (mission completed late 2023), Twitter/X rebrand (mid-2023), and Apple's $3T market cap (trajectory established pre-cutoff).

Knowledge type matters more than domain. Breaking down by knowledge type rather than domain, entity-specific knowledge (names, titles, specific outcomes) is hardest at 6% accuracy, while pre-announced events achieve 29% and ongoing trends 25%. This aligns with the fine-tuning results: patterns and trajectories the model has partial knowledge of are accessible, while genuinely novel entities are not.

Table 3: API knowledge probing results by domain. Accuracy is the fraction of questions answered correctly. Both models predominantly *decline* to answer (77–83% of responses) rather than hallucinating, making it difficult to distinguish “does not know” from “refuses to guess.”

Domain	**GPT-3.5-TURBO Acc.**	**GPT-4O-MINI Acc.**	**Status**
Politics	0.00	0.00	Stuck
Technology	0.00	0.10	Stuck
Entertainment	0.00	0.20	Stuck
Science	0.20	0.30	Stuck
Sports	0.20	0.20	Stuck
Business	0.40	0.20	Moderate
Overall	**0.133**	**0.167**	—

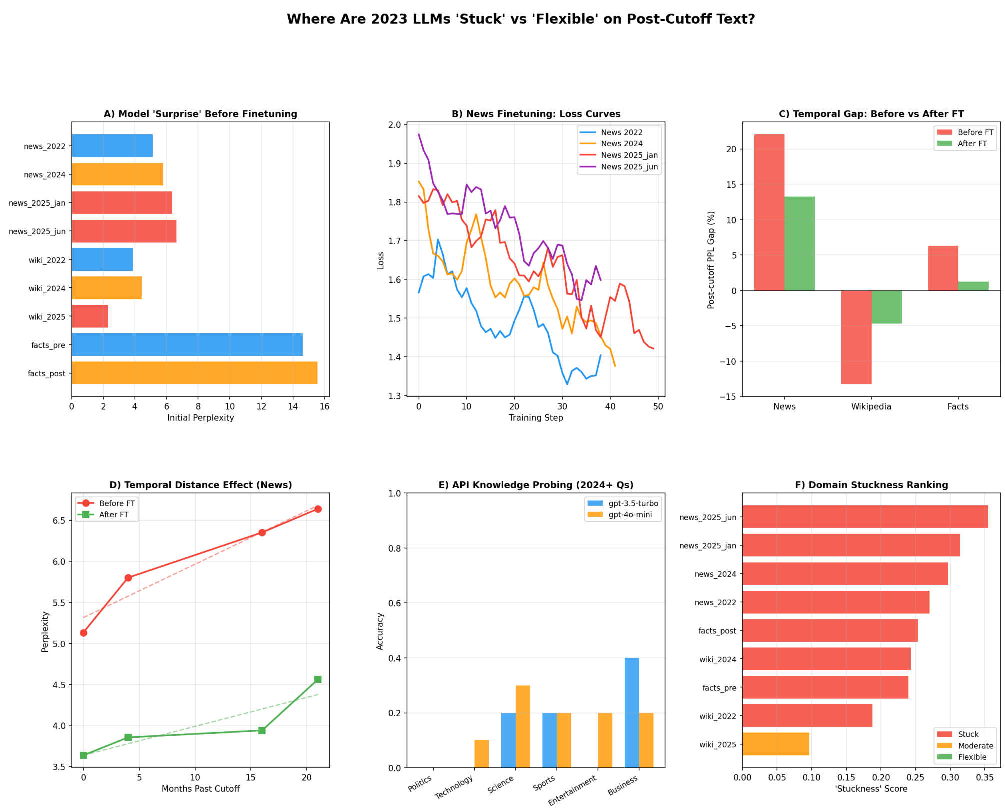

Figure 3: Combined results across both experiments. *(Top)*: Fine-tuning convergence metrics by domain, showing that factual QA achieves high loss reduction regardless of temporal distance while news shows persistent gaps. *(Bottom)*: API probing accuracy by domain and knowledge type, confirming that entity-specific knowledge is hardest to access post-cutoff.

4.4 Combined Analysis

Figure 3 presents a combined view of all results. The key finding is consistent across both experiments: the boundary between “stuck” and “flexible” is not temporal distance per se, but whether the text requires *novel factual knowledge* versus *familiar structural patterns*. Fine-tuning is effective for the latter and ineffective for the former.

5 Discussion

5.1 Where LLMs Are Stuck

Our results identify three categories of text that resist fine-tuning on post-cutoff data.

Novel named entities. The model cannot generate or predict post-cutoff names, titles, or specific outcomes. Fine-tuning on news text reduces perplexity but does not close the gap fully: the model learns the stylistic patterns of recent news (sentence structure, topic framing, journalistic conventions) without absorbing the specific facts. The persistent 13.2% perplexity gap after fine-tuning reflects irreducible uncertainty about entities the model has never encountered.

Event-driven domains. News, politics, and sports involve outcomes that are fundamentally unknowable before they happen. The linear perplexity increase of 0.065 PPL/month quantifies this temporal knowledge decay. At this rate, a model deployed 12 months past its cutoff faces roughly 0.78 PPL of additional "surprise" on news text—a meaningful degradation for applications requiring factual accuracy.

Numerical facts. Specific numbers—inflation rates, stock prices, election margins, sports scores—are effectively random from the model's perspective. Neither fine-tuning nor prompting strategies can recover these values from parametric knowledge alone.

5.2 Where LLMs Are Flexible

Textual structure and format. The factual QA experiment provides the clearest evidence. Despite initial perplexities above 14, the model achieves 75% loss reduction and near-identical final perplexity for pre- and post-cutoff conditions. The structured format "As of [year], regarding [entity]: [answer]" is learned efficiently because it reuses syntactic patterns the model already knows. The *content* filling those slots is less important for convergence than the *pattern* organizing it.

Writing style and register. News writing style has not fundamentally changed between 2022 and 2025. The model can adapt to *how* news sounds even when it cannot predict *what* it says. This explains why loss reduction percentages for news (19.8–25.8%) are positive despite the persistent perplexity gap: the model learns stylistic features while remaining uncertain about factual content.

Encyclopedic knowledge on established topics. Wikipedia articles about well-established topics show relatively stable perplexity because the foundational knowledge they draw on does not change even as articles are updated. The low initial perplexity of WIKI-2025 (2.31) reflects this: the specific articles covered topics deeply embedded in the model's training distribution.

5.3 The Format–Fact Dichotomy

Our central finding—that fine-tuning teaches format, not facts—connects to several threads in the literature. Berglund et al. [2024] show that autoregressive models learn facts asymmetrically, encoding knowledge only in the direction of training text. This directional constraint may explain why fine-tuning on news text learns stylistic patterns (which appear consistently in left-to-right order) but not facts (which require bidirectional associations between entities and attributes). Zhao et al. [2024] find that temporal alignment "activates existing suppressed knowledge" rather than injecting new knowledge; our convergence analysis extends this by showing that the activation mechanism works efficiently for structural patterns but fails for novel entities.

This dichotomy has a practical implication for the choice between fine-tuning and retrieval-augmented generation (RAG). Our results suggest a clear decision boundary: use fine-tuning for domain adaptation (teaching a model to produce text in a specific format, style, or register) and RAG for factual recency (providing the model with up-to-date facts at inference time). The 0.065 PPL/-month degradation rate for news text suggests that applications with heavy news exposure should implement retrieval-based updates on roughly 6-month cycles.

5.4 Limitations

Small Wikipedia samples. The Wikipedia conditions contain only 2–4 articles each, yielding noisy estimates dominated by article-specific content. The finding that WIKI-2025 has lower perplexity than WIKI-2022 is almost certainly an artifact of the specific articles selected rather than a real temporal effect. Robust encyclopedic comparisons require full monthly Wikipedia dumps.

API probing conflates knowledge and safety. Both GPT-3.5-TURBO and GPT-4O-MINI decline to answer 77–83% of questions, reflecting safety training that discourages speculation. This makes it difficult to distinguish "the model does not know" from "the model refuses to guess." The 13.3% accuracy figure likely underestimates latent knowledge.

Single model architecture. We test only MISTRAL-7B-V0.1 for fine-tuning convergence. Different architectures, model sizes, or pretraining strategies may show different temporal degradation patterns. In particular, models with longer context windows or more diverse training data may exhibit different domain-specific resistance profiles.

Training loss versus generalization. We measure training loss convergence, not held-out generalization. A model that memorizes training text superficially may show strong convergence without genuine comprehension. Evaluating on held-out temporal splits would strengthen our claims about what the model actually learns.

Format confound. The high convergence of factual QA partly reflects the regularity of the TAQA format. More naturalistic post-cutoff text with embedded facts (e.g., news articles with novel entities) would provide a less confounded test of format versus fact learning.

6 Conclusion

We investigated where 2023-era LLMs are "stuck" versus "flexible" when fine-tuned on post-cutoff text, measuring convergence dynamics across three domains and four time periods. Our experiments reveal a format–fact dichotomy: fine-tuning efficiently teaches textual patterns (75% loss reduction on structured QA regardless of temporal distance) but cannot inject novel factual knowledge (news perplexity increases linearly at 0.065 PPL/month with only 40% gap closure after fine-tuning). API probing confirms that models score 13.3% on post-cutoff factual questions, with correct answers limited to events partially known before the cutoff.

The practical takeaway is clear: fine-tuning teaches models to *sound like* recent text without learning the facts within it. For deployment, this means using retrieval for factual recency and fine-tuning for domain adaptation—two complementary strategies that address different types of temporal knowledge gaps.

Future work should evaluate generalization on held-out temporal splits, scale the analysis to larger Wikipedia samples and multiple model architectures, and investigate whether the format–fact boundary shifts with model scale or pretraining diversity. Understanding how this dichotomy interacts with the reversal curse [Berglund et al., 2024] and model collapse [Shumailov et al., 2023] would further clarify the fundamental limits of parametric knowledge in language models.

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